Analisis **IndoBERT** untuk **13 Bank Digital** Indonesia

1. Introduction

Analisis aspect category, opinion, sentiment, dan emotion dengan IndoBERT sangat relevan untuk 13 bank digital Indonesia, tetapi bukti yang tersedia saat ini lebih kuat untuk **kelayakan teknis klasifikasi** daripada untuk **inferensi bisnis lintas-bank** atau **pemetaan emosi yang tervalidasi**. Literatur menunjukkan bahwa ulasan aplikasi dan percakapan media sosial memang menangkap kepuasan, keluhan teknis, kualitas layanan, dan kepercayaan pengguna, sementara pendekatan berbasis IndoBERT umumnya mengungguli model klasik untuk klasifikasi sentimen dan banyak tugas ABSA (aspect-based sentiment analysis) berbahasa Indonesia (Yulianti & Nissa, 2024; Pardamean et al., 2025; Dhendra, 2025; Casanova et al., 2026; Saputra et al., 2026; Jazuli et al., 2024).

Namun, korpus ini juga menunjukkan tiga batas besar. Pertama, studi perbankan Indonesia yang benar-benar mencakup banyak bank masih sedikit, dan yang ada biasanya hanya menilai 1–3 aplikasi atau satu platform data, bukan 13 bank digital sekaligus (Andrian et al., 2022; Syaripudin et al., 2025; Tomasaputra & Siallagan, 2025; Zahratussyafitri et al., 2026). Kedua, banyak studi memberi label sentimen dari rating bintang atau pseudo-label otomatis, sehingga hubungan antara opinion, sentiment, emotion, dan kepuasan dapat sebagian “dibangun” oleh desain anotasi, bukan sepenuhnya ditemukan dari data alami (Casanova et al., 2026; Setyawati et al., 2026; Maretta & Meiriza, 2025; Hazz & Wahyuningsih, 2026). Ketiga, generalisasi lintas domain, lintas waktu, dan lintas platform masih menjadi masalah terbuka, terutama untuk teks Indonesia yang informal, bercampur kode, tidak seimbang kelasnya, dan sering memuat multi-aspek dalam satu ulasan (Susanto et al., 2026; Nataprawira & Nurhaida, 2026; Mozetič et al., 2018; Al-Moslmi et al., 2017).

Apakah analisis aspek, opini, sentimen, dan emosi dengan IndoBERT relevan untuk mengevaluasi 13 bank digital Indonesia?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Konsensus relevansi IndoBERT untuk evaluasi bank digital

2. Methods

Tinjauan ini mensintesis korpus Deep Search Consensus atas literatur tentang sentiment analysis, aspect-based sentiment analysis, klasifikasi emosi, dan kualitas layanan pada perbankan digital Indonesia serta domain Indonesia yang berdekatan. Pencarian dijalankan di atas lebih dari 220 juta paper dalam Consensus, yang mengindeks Semantic Scholar, PubMed, dan sumber ilmiah lain, lalu dipersempit ke studi yang relevan dengan IndoBERT, ulasan aplikasi bank, evaluasi lintas-bank, pelabelan sentimen/emosi, serta isu validitas seperti class imbalance, domain shift, dan privasi.

Search Strategy

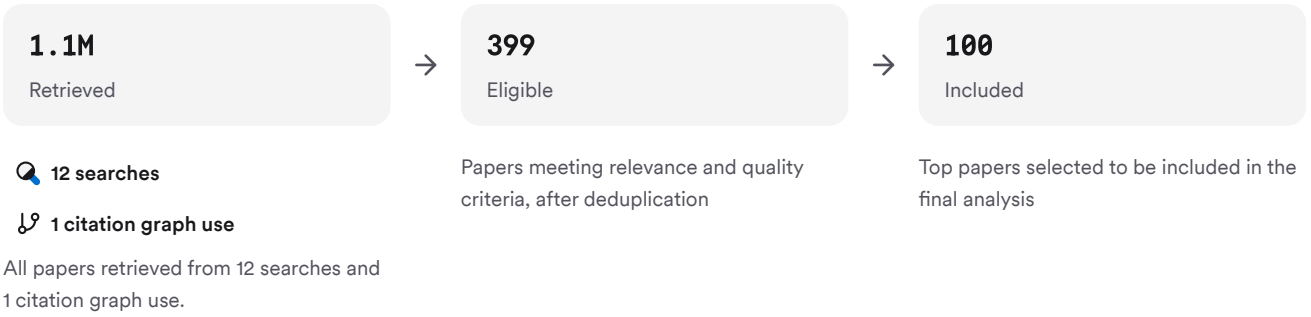


FIGURE 2 Strategi pencarian dan penyaringan studi

Strategi menggabungkan pencarian topik bank digital Indonesia, benchmark IndoBERT, studi ABSA, dan paper validitas evaluasi lintas-domain.

3. Results

3.1 Key Papers

Fondasi korpus ini bertumpu pada empat jenis studi: studi kepuasan bank digital Indonesia lintas merek, studi ABSA IndoBERT berbahasa Indonesia, studi kualitas layanan mobile banking, dan studi metodologi evaluasi model pada data temporal atau lintas-domain (Andrian et al., 2022; Yulianti & Nissa, 2024; Tomasaputra & Siallagan, 2025; Mozetič et al., 2018).

Paper	Summary
(Andrian et al., 2022)	Tiga bank digital, distribusi sentimen berbeda (Andrian et al., 2022)
(Yulianti & Nissa, 2024)	IndoBERT unggul untuk ABSA Indonesia (Yulianti & Nissa, 2024)
(Tomasaputra & Siallagan, 2025)	Sentimen dipetakan ke dimensi E-S-QUAL (Tomasaputra & Siallagan, 2025)

FIGURE 3 Paper jangkar dalam korpus ini

3.2 Performa Model

Sebagian besar studi Indonesia menemukan IndoBERT kuat untuk sentimen dan ABSA, sering mengungguli mBERT, SVM, CNN, BiLSTM, atau XLM-R pada data ulasan dan layanan berbahasa Indonesia (Doewes & Widoyono, 2026; Citra & Ghifari, 2026; Doewes & Widoyono, 2026; Dhendra, 2025; Sihombing et al., 2026). Pada domain perbankan atau fintech yang dekat, IndoBERT mencapai akurasi sekitar 87% pada ulasan blu, 88% pada mobile banking syariah, 90% pada BRImo, 91.24% pada DANA, 94%–95% pada beberapa tugas bank/dompet digital, dan F1 0.9373 pada knowledge-discovery pipeline bank (Casanova et al., 2026; Assani et al., 2025; Simarmata & Sasongko, 2025; Susanto et al., 2026; Sanjaya et al., 2025; Saputra et al., 2026).

Tetapi keunggulan IndoBERT tidak universal. Pada data kecil atau setup tidak seimbang, mBERT atau linear SVC kadang unggul, terutama bila ada sinyal tambahan seperti rating atau sampling yang tidak setara antar model (Afandy et al., 2026; Zahra et al., 2026; Kadarsih & Pujianto, 2025).

3.3 Temuan Substantif Perbankan

Di studi perbankan Indonesia, sinyal yang paling konsisten adalah bahwa ulasan negatif menumpuk pada **reliability**: login, verifikasi, crash, error transaksi, performa lambat, dan respons customer service yang lemah (Prasetyo & Agastya, 2024; Simarmata & Sasongko, 2025; Tania & Afrina, 2026; Zahratussyafitri et al., 2026; Arif et al., 2026; Adiningtyas & Auliani, 2024).

Sebaliknya, ulasan positif biasanya terkait efisiensi transaksi, kemudahan penggunaan, kenyamanan, fitur, dan antarmuka (Casanova et al., 2026; Prasetyo & Agastya, 2024; Tomasaputra & Siallagan, 2025; Arif et al., 2026; Oh & Kim, 2022).

Perbandingan lintas bank juga menunjukkan heterogenitas besar, sehingga studi 13 bank tidak boleh mengasumsikan pola sentimen yang seragam. Dalam korpus Indonesia, Jago pernah dominan positif di Twitter 2021, tetapi pada ulasan aplikasi periode lain bisa dominan negatif; SeaBank tampak unggul pada beberapa studi review-store, sedangkan Blu, Livin', Neo, dan bank lain menunjukkan pola yang lebih problematik atau bercampur tergantung platform dan waktu (Andrian et al., 2022; Syaripudin et al., 2025; Tomasaputra & Siallagan, 2025; Zahratussyafitri et al., 2026; Mulyanto et al., 2025).

3.4 Opinion Dan Emotion

ABSA jelas lebih informatif daripada polaritas dokumen tunggal karena satu ulasan sering memuat beberapa aspek dan sentimen campuran (Pardamean et al., 2025; Wirawan et al., 2026; Halik et al., 2026; Hayatin et al., 2024). Literatur juga menunjukkan bahwa level yang lebih kaya—opinion term, aspect category, bahkan quadruple extraction—secara konseptual lebih cocok bila tujuan studi 13 bank adalah mengetahui *aspek apa*, *opini apa*, dan *sentimen apa* yang muncul bersama, termasuk aspek atau opini implisit (Cai et al., 2021; Zhang et al., 2024; Zhang et al., 2022).

Bukti untuk analisis emosi ada, tetapi lebih tipis di konteks perbankan Indonesia. IndoBERT dapat mengklasifikasi emosi Indonesia dengan F1 sekitar 0.76–0.79 pada benchmark tweet dan hingga 0.93 pada tugas multi-label tertentu (Ahmadian et al., 2023; Shaw et al., 2025; Amalia et al., 2025). Dalam layanan finansial, emosi seperti trust, joy, anticipation, anger, fear, dan sadness memang muncul dan berhubungan dengan persepsi layanan, tetapi sebagian studi emosi perbankan masih berbasis leksikon, bukan model bank-Indonesia yang tervalidasi secara khusus (Omotosho, 2021; Perea-Khalifi et al., 2024; Dhar & Bose, 2022).

Results Timeline

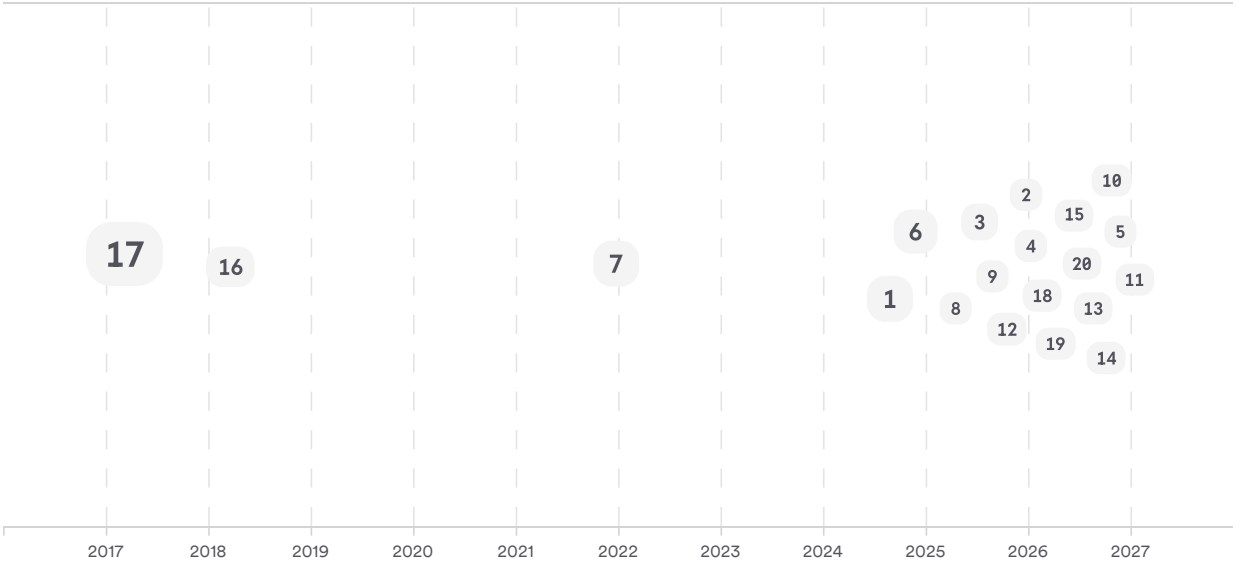


FIGURE 4 Linimasa hasil riset, marker besar lebih tersitasi

Top Contributors

Type	Name	Papers
Author	Hendri Ahmadian	(Ahmadian et al., 2024; Ahmadian et al., 2023)
Author	Ken Ditha Tania	(Casanova et al., 2026; Tania & Afrina, 2026; Fatihaturrahmah & Tania, 2025)
Author	Allsela Meiriza	(Casanova et al., 2026; Maretta & Meiriza, 2025)
Journal	*Journal of Applied Informatics and Computing*	(Casanova et al., 2026; Maretta & Meiriza, 2025; Simarmata & Sasongko, 2025; Tania & Afrina, 2026)
Journal	*International Journal of Bank Marketing*	(Ghadiridehkordi et al., 2024; Çallı, 2022)
Journal	*Neural Computing and Applications*	(Mahmood et al., 2023; Alrizq & Alghamdi, 2024)

FIGURE 5 Penulis dan jurnal paling sering muncul

4. Discussion

Secara metodologis, analisis ini sangat relevan untuk 13 bank digital karena langsung memetakan suara pelanggan ke dimensi layanan yang bisa ditindaklanjuti, terutama reliability, responsiveness, efficiency, usability, security, dan trust (Tomasaputra & Siallagan, 2025; Oh & Kim, 2022; Al-Kamal et al., 2025; Da Costa Martins, 2026). Korelasinya juga masuk akal: sentimen dan topik tertentu berkaitan dengan rating, kepuasan, kepercayaan, dan niat adopsi, dengan keamanan, kemudahan, dan kualitas layanan berulang kali muncul sebagai determinan utama (Dhar & Bose, 2022; Oh & Kim, 2022; Desiraju et al., 2024; Kantika et al., 2022).

Namun, bukti kausal tentang “dampak” masih terbatas. Sebagian besar paper menunjukkan bahwa analisis sentimen/ABSA membantu **mengidentifikasi** prioritas perbaikan, bukan bahwa penerapannya terbukti **menyebabkan** retensi, loyalitas, atau perbaikan performa bank di dunia nyata (Casanova et al., 2026; Tania & Afrina, 2026; Leem & Eum, 2021; Albesher & Alsanousi, 2026). Ini penting karena banyak studi berhenti pada klasifikasi dan interpretasi, tanpa evaluasi intervensi sesudah bank memperbaiki fitur tertentu.

Masalah validitas juga nyata. Banyak dataset memakai weak labels dari rating, pseudo-label otomatis, atau model lain, yang bisa menambah noise dan circularity; neutral dan mixed sentiment juga tetap paling sulit; dan evaluasi acak pada data temporal dapat melebihkan performa dibanding skenario produksi nyata (Setyawati et al., 2026; Bimasena et al., 2026; Perwira et al., 2025; Maretta & Meiriza, 2025; Mozetič et al., 2018; Hazz & Wahyuningsih, 2026).

Akhirnya, studi 13 bank juga harus memperhitungkan etika dan privasi. Ulasan publik tetap membawa risiko re-identifikasi, dan anonimisasi sederhana tidak cukup; praktik yang lebih aman memerlukan minimisasi bahaya, transparansi, dan audit risiko privasi (Ford et al., 2021; Gadotti et al., 2024; Manzanares-Salor et al., 2024; Shahriar et al., 2025).

Claim	Evidence Strength	Reasoning	Papers
IndoBERT efektif untuk sentimen/ABSA bahasa Indonesia	Strong	Didukung banyak benchmark dan domain Indonesia	(Yulianti & Nissa, 2024; Doewes & Widoyono, 2026; Dhendra, 2025; Sihombing et al., 2026)
Keluhan bank digital terkonsentrasi pada reliability dan support	Strong	Konsisten lintas studi bank, e-wallet, dan kualitas layanan	(Simarmata & Sasongko, 2025; Tania & Afrina, 2026; Zahratussyafitri et al., 2026; Adiningtyas & Auliani, 2024)
ABSA lebih berguna daripada polaritas tunggal untuk 13 bank	Moderate	Ulasan multi-aspek dan mixed sentiment berulang kali dilaporkan	(Pardamean et al., 2025; Wirawan et al., 2026; Halik et al., 2026)
Analisis emosi menambah sinyal di luar sentimen	Moderate	Ada bukti teknis dan konseptual, tetapi validasi perbankan Indonesia masih terbatas	(Amalia et al., 2025; Omotosho, 2021; Dhar & Bose, 2022)

Claim	Evidence Strength	Reasoning	Papers
Dampak bisnis kausal langsung sudah terbukti kuat	Weak	Mayoritas studi observasional tanpa evaluasi intervensi	(Casanova et al., 2026; Albesher & Alsanousi, 2026; Alrizq & Alghamdi, 2024)

FIGURE 6 Klaim kunci dan kekuatan evidensinya

5. Conclusion

Secara keseluruhan, analisis aspect category, opinion, sentiment, dan emotion dengan IndoBERT relevan untuk menilai 13 bank digital Indonesia karena dapat mengubah ulasan mentah menjadi peta masalah layanan yang spesifik. Korelasinya dengan kepuasan, trust, rating, dan kualitas layanan tampak cukup konsisten, tetapi dampak kausal pada hasil bisnis masih belum mapan dan desain studi lintas-13-bank yang kuat belum tersedia.

Research Gaps

Literatur saat ini padat pada klasifikasi sentimen aplikasi tunggal atau 2–3 bank, tetapi tipis pada evaluasi komparatif 13 bank dengan anotasi manusia yang kaya. Celah terbesar bukan lagi “apakah IndoBERT bisa bekerja,” melainkan “apakah hasilnya valid, sebanding lintas bank, dan cukup rinci untuk keputusan operasional.”

	Google Play	Twitter X	Manual Label	Cross Bank	Intervensi
Sentimen Bank	29 papers	8 papers	12 papers	2 papers	No papers Potential gap
ABSA Bank	7 papers	1 paper	6 papers	1 paper	No papers Potential gap
Emosi Bank	4 papers	3 papers	3 papers	No papers Potential gap	No papers Potential gap
Trust Adoption	1 paper	No papers Potential gap	2 papers	No papers Potential gap	No papers Potential gap
Privasi Teks	No papers Potential gap	No papers Potential gap	No papers Potential gap	No papers Potential gap	No papers Potential gap

Studi lintas-bank besar masih jarang, dan kebanyakan hanya mencakup satu aplikasi atau tiga bank (Syaripudin et al., 2025; Tomasaputra & Siallagan, 2025; Andrian et al., 2022). Studi emosi juga tertinggal jauh dibanding sentimen dan ABSA pada konteks bank Indonesia, meski emosi seperti trust, anger, dan joy tampak relevan secara teoritis dan empiris di domain finansial lain (Omotosho, 2021; Perea-Khalifi et al., 2024; Amalia et al., 2025). Kesenjangan lain adalah kurangnya validasi temporal, lintas-platform, dan privasi-terjaga, padahal domain shift dan risiko re-identifikasi sudah jelas terdokumentasi (Mozetič et al., 2018; Fatihaturrahmah & Tania, 2025; Manzanares-Salor et al., 2024).

Open Research Questions

Pertanyaan berikut paling bernilai untuk studi lanjutan pada 13 bank digital Indonesia.

Question	Why
Apakah model IndoBERT yang dilatih pada satu bank tetap akurat ketika diuji pada 12 bank digital Indonesia lain?	Ini menguji generalisasi lintas-bank, bukan hanya kecocokan pada satu korpus pelatihan.
Apakah aspect-category-opinion-sentiment quadruple extraction memberi wawasan operasional yang lebih baik daripada ABSA biasa pada ulasan bank digital Indonesia?	Ini dapat menangkap opini implisit dan hubungan aspek-opini yang hilang pada label polaritas sederhana.
Apakah perbaikan pada aspek reliability yang diidentifikasi model benar-benar menurunkan keluhan dan menaikkan kepuasan pada periode berikutnya?	Ini mengubah literatur dari deteksi deskriptif menjadi evaluasi dampak kausal yang praktis.

Tinjauan ini menunjukkan bahwa IndoBERT sudah cukup matang untuk studi 13 bank digital Indonesia, tetapi nilai ilmiah tertinggi akan datang dari desain yang ketat, anotasi valid, dan evaluasi lintas-bank yang benar-benar adil.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at https://consensus.app. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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